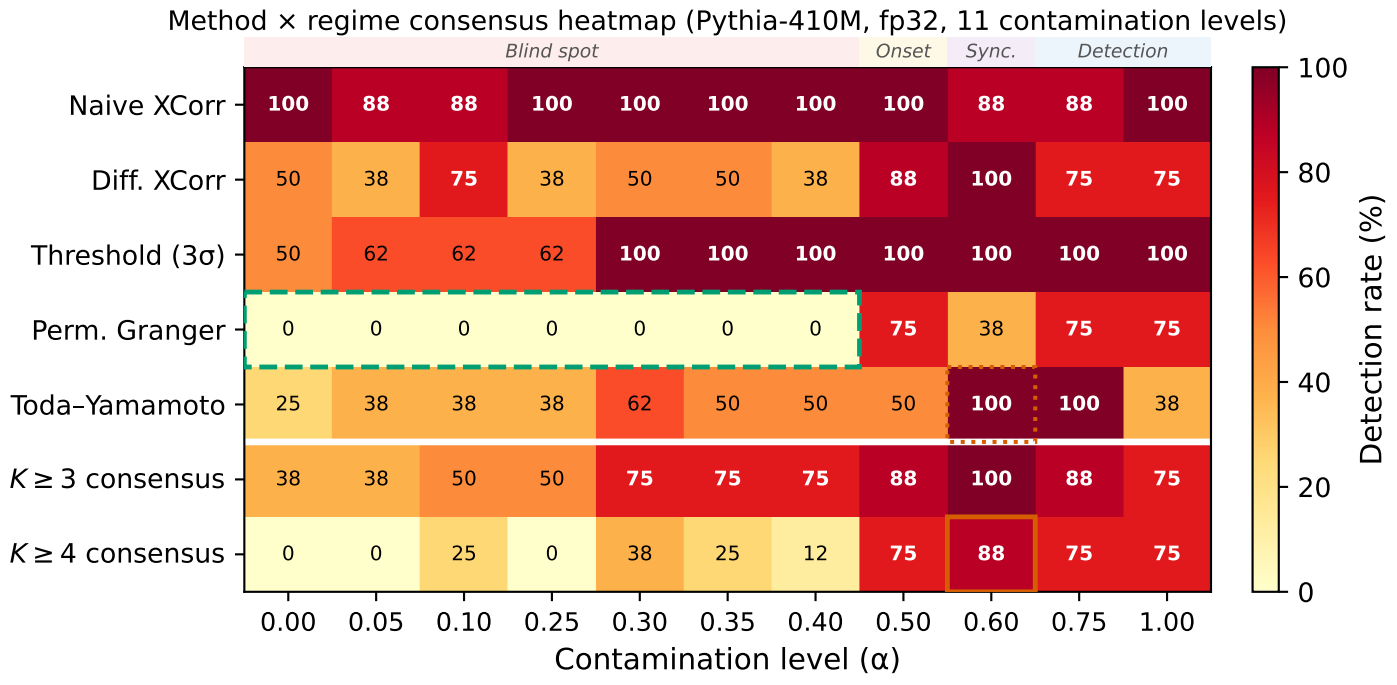
Method × regime consensus heatmap (Pythia-410M, fp32, 11 contamination levels)
Blind spot
Onset
Sync.
Detection
Naive XCorr
Diff. XCorr
Threshold (3σ)
Perm. Granger
Toda–Yamamoto
K ≥ 3 consensus
K ≥ 4 consensus
100 88 88 100 100 100 100 100 88 88 100
50 38 75 38 50 50 38 88 100 75 75
50 62 62 62 100 100 100 100 100 100 100
0 0 0 0 0 0 0 75 38 75 75
25 38 38 38 62 50 50 50 100 100 38
38 38 50 50 75 75 75 88 100 88 75
0 0 25 0 38 25 12 75 88 75 75
0.00 0.05 0.10 0.25 0.30 0.35 0.40 0.50 0.60 0.75 1.00
Contamination level (α)
Detection rate (%)
100
80
60
40
20
0